

Course Overview

This beginner-level Data Structures and Algorithms (DSA) course is designed to build strong problem-solving skills and programming fundamentals. Students will learn core data structures, algorithmic thinking, and coding techniques commonly used in technical interviews and real-world software development.

Module 1: Programming Fundamentals

- Introduction to Programming Logic
- Variables, Data Types, and Operators
- Conditional Statements
- Loops and Iteration
- Functions and Recursion Basics

Module 2: Time & Space Complexity

- Why Complexity Matters
- Big-O Notation
- Best, Average, and Worst Case Analysis
- Common Complexity Patterns

Module 3: Arrays

- Array Basics
- Traversal & Searching
- Insertion & Deletion
- Two Pointer Technique
- Sliding Window Technique

Module 4: Strings

- String Manipulation
- Pattern Matching Basics
- Anagrams & Palindromes
- String Optimization Problems

Module 5: Recursion & Backtracking

- Recursion Concepts
- Call Stack Understanding
- Backtracking Fundamentals
- Classic Problems (Subsets, Permutations)

Module 6: Linked Lists

- Singly Linked List
- Doubly Linked List
- Insertion & Deletion
- Cycle Detection
- Reversal Techniques

Module 7: Stacks & Queues

- Stack Implementation
- Queue Implementation
- Applications of Stack
- Circular Queue & Deque
- Monotonic Stack Basics

Module 8: Trees

- Binary Trees
- Tree Traversals (DFS & BFS)
- Binary Search Trees
- Tree Height & Depth
- Basic Tree Problems

Module 9: Heaps & Priority Queues

- Heap Concepts
- Min Heap & Max Heap
- Heap Operations
- Priority Queue Applications

Module 10: Hashing

- Hash Tables
- Collision Handling
- Hash Maps & Sets

- Frequency Counting Problems

Module 11: Graph Basics

- Graph Representation
- BFS & DFS
- Connected Components
- Shortest Path Basics

Module 12: Sorting & Searching Algorithms

- Linear & Binary Search
- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort (Intro)

Module 13: Greedy & Basic Dynamic Programming

- Greedy Strategy Basics
- Intro to Dynamic Programming
- Memoization vs Tabulation
- Classic Beginner DP Problems

Module 14: Interview Preparation

- Problem-Solving Strategies
- Coding Patterns
- Mock Interview Practice
- Debugging Techniques

Learning Outcomes

- Understand fundamental data structures
- Analyze algorithm efficiency using Big-O notation
- Solve coding problems using structured approaches
- Prepare for technical coding interviews
- Build strong logical and analytical thinking skills